

Data Cleaning Log Book: E-Commerce Sales Dataset

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Project Overview

I was given a raw e-commerce dataset with 15 columns and over 299 rows. The goal was to clean the data using Power Query, so it would be ready for analysis, pivot tables, and dashboard creation. I followed a strict workflow: Understand → Inspect → Clean → Validate.

Phase 1: Understand (Know Your Data)

Before touching anything, I looked at every column to understand what it represented. I asked myself: *"Will I need this to analyze sales?"*

Initial Columns: RowNum, OrderID, CustomerName, Email, Phone, Product, Category, Quantity, Unit Price, OrderDate, ShipDate, Country, PaymentMethod, OrderStatus, Notes

My Decisions:

- I kept OrderID, CustomerName, Product, Category, Quantity, Unit Price, OrderDate, ShipDate, Country, PaymentMethod, and OrderStatus. These are essential for sales analysis.
- I removed Email and Phone because they aren't needed for sales calculations and removing them protects customer privacy.
- I removed Notes because it was mostly empty, with only a few entries like "VIP customer," which wasn't reliable or structured enough for analysis.
- I removed RowNum because it was just a useless row counter.

Phase 2: Inspect (Find the Mess)

This was the most critical phase. I systematically checked the data for problems.

What I found:

- **Inconsistent Capitalization:** Categories like beauty, BEAUTY, and Beauty all meant the same thing. Customer names and statuses had the same issue.
- **Extra Spaces:** Some cells had invisible spaces that could break matching later.

- **Duplicate Records (The tricky part):** I found duplicates in OrderID. However, I realized that OrderID is a group identifier (one order can have multiple products), not a unique row identifier. I knew that if I just deleted duplicates based on OrderID, I would destroy the data.
- **Quantity Column Mess:** I noticed rows with -1, -3, -2. Some of these were **Returned** orders (correct), but some were **Processing, In Progress, Delivered, and Shipped** (which is a typo).
- **Unit Price Column Mess:** The values were a disaster. Some had NGN, some had \$ signs, some had commas (e.g., 28,700), some were negative, and some were null.
- **Date Column Mess:** Dates were mixed. Some were American (MM/DD/YYYY) and some were British (DD/MM/YYYY), causing errors in Power Query.
- **Null Values Across Columns:** I found several null values spread across Quantity, Unit Price, OrderDate, and ShipDate.

Phase 3: Clean (Fix the Problems)

I tackled the problems in a logical order. I fixed text first, then numbers, then dates.

1. Text Standardization (Capitalization & Trimming)

- **Action:** I selected text columns like CustomerName, Product, Category, Country, PaymentMethod, and OrderStatus. I used **Trim** to remove hidden spaces and **Capitalize Each Word** to fix the case.
- **Result:** beauty became Beauty, diane sanchez became Diane Sanchez. This ensures Pivot Tables group correctly.

2. Removing True Duplicates (The Critical Logic)

- **The Trap:** If I selected just the OrderID column and clicked "Remove Duplicates," I would have deleted legitimate products because one order can have many rows (e.g., Order #100 has a Laptop and a Mouse).
- **My Fix:** I selected **all columns EXCEPT** OrderID (and I included the OrderDate to keep separate purchases safe). I then did **Remove Duplicates**.
- **Why this works:** Removing duplicates on the actual details (Customer, Product, Quantity, Date) safely removes only the true typing errors, while keeping multi-product orders perfectly intact.

3. Fixing the "Quantity" Column (Negative Typos vs. Real Returns)

- **The Problem:** I noticed rows with -1, -3, -2. Some were **Returned** (which is correct, returns reduce stock). However, some were **Delivered, Shipped, In Progress, or Processing**. A delivered order cannot have a negative quantity; it was clearly a **data entry typo**.

- **My Fix:** I created a custom column with this logic: *If the status is "Returned", keep the negative number as it is (it's correct). If the quantity is less than 0 (and it's NOT a return), multiply it by -1 to make it positive (e.g., changing -3 delivered to +3). If the quantity is empty (null), leave it as null.*
- **Result:** Returns correctly subtract from sales and shipped/delivered typos are fixed.

4. Tracking Data Quality in "Quantity"

- I created a separate column called Quantity_Quality_Flag. This column labels each row as either "Valid" or "Missing Quantity".
- I did this, so I can easily count how many null values exist and measure the overall quality of the Quantity column without destroying the numeric data type.

5. Fixing the "Unit Price" Column

- **The Problem:** The column was a mess. I found negative values, null, values with NGN, and values with \$ signs. The prices should be in Naira, but the symbols prevented it from being a proper number.
- **My Fix:** I used a custom formula to strip out all the text (letters, symbols, commas) and keep only the raw numbers. Then, I applied a second rule: since the Unit Price is the cost of *one* item, it can never be negative. I flipped any remaining negative prices into positive numbers.
- **Result:** A clean numeric column ready for revenue calculations.

6. Fixing the "Date" Columns

- **The Problem:** Power Query was throwing errors because some dates were American (12/13/2025) and some were British (22/03/2026).
- **My Fix:** I created a custom column that tries to read the date as American format first. If that fails (e.g., Month 22 doesn't exist), it tries British format. If both fail, it leaves the date as null.
- **Result:** All readable dates were converted into a single, proper Date data type.

Phase 4: Validate (Check My Work and Business Questions)

I ran several checks before loading to ensure my fixes didn't break anything.

- **Row Count & Column Count:** I verified the counts were logical. I only lost a few rows to true duplicates. The columns I removed (Email, Phone, Notes, RowNum) were gone.
- **Text Categories:** I clicked the filter on Category and OrderStatus. I only saw clean, consistent values like Beauty, Electronics, Delivered, and Returned. No more mixed cases.

- **Duplicates Check:** I ran **Keep Duplicates** again on my key columns. Zero rows appeared, confirming the true duplicates were gone.
- **Logical Spot Checks:** I found a delivered order that originally had -3. After cleaning, it now shows +3. The math makes sense. I found a returned order. The Quantity stayed negative (-1), and the Price is positive. Multiplying them gives a negative revenue (correct for a refund).

The Most Interesting Discovery (Business Question Found!)

While validating the duplicates, I spotted something fascinating.

The Case of Order ID ORD-100142: It appeared twice, both had the **same Ship Date**. However, one row had a Quantity of 5 and was **Delivered**.

The other row had a **different Unit Price** and was **Cancelled**.

This is not a duplicate error. This is a genuine business question.

Why was part of the order cancelled?

Why does the cancelled part have a different unit price than the delivered part?

This is exactly the kind of insight I want to take back to the sales department for qualitative analysis. My cleaning process successfully preserved this nuance, so the business team can investigate it.

Root Cause Analysis & Recommendations for the Business

During the inspection phase, I noticed several recurring issues that caused the mess in the first place. I am recommending systemic fixes to prevent these errors from happening in future data entries.

1. **Why do we have** null values in Quantity, Unit Price, OrderDate, and ShipDate?

- **Root Cause:** These are likely due to human error during data entry (the person simply skipped the field) or system integration failures (e.g., the shipping system didn't send the date back to the main database).
- **Impact:** null values make our reports inaccurate. For example, if a delivered order has a null Quantity, we cannot calculate its revenue.

2. **Why do we have inconsistent spellings in** Category, Country, and PaymentMethod?

- **Root Cause:** Data entry clerks are typing these fields manually in a free-text box. This leads to beauty, BEAUTY, and Beauty all being entered for the same product type.

- **Impact:** This destroys Pivot Table grouping. The sales team might think they have three different categories when they are all the same.

Recommended Technical Solutions:

A. Implement Pre-Defined Dropdown Menus (Validation Lists)

- **Action:** Change the data entry system so that fields like Category, Country, PaymentMethod, and OrderStatus are selected from a dropdown list rather than typed manually.
- **Benefit:** This completely eliminates capitalization errors and misspellings. Every entry will be exactly Beauty or Electronics every time.

B. Enforce Mandatory Fields (Required Entry)

- **Action:** Set Quantity, Unit Price, and OrderDate as **required fields** in the system. The system should not allow the user to save the order unless these fields are filled in.
- **Benefit:** This drastically reduces the number of null values we have to deal with in our analysis.

C. Use a Date Picker (Calendar Widget)

- **Action:** Replace the free-text date box with a pop-up calendar for OrderDate and ShipDate.
- **Benefit:** This ensures all dates are saved in a single, unified format (e.g., YYYY-MM-DD), eliminating the confusion between American and British formats.